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Some Quantum-Mechanical Considerations in the Theory of Reactions Involving an Activation Energy

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The activated complex or transition state method for calculating the absolute rate of a chemical reaction with an activation energy would be rigorously valid if classical mechanics applied to all degrees of freedom. In quantum mechanics, two kinds of limitations must be considered. First, because of Heisenberg's uncertainty principle, the transition state itself can be defined only if the potential surface is sufficiently flat around the highest point of the reaction path. Second, even if this condition is fulfilled, the transmission coefficient can differ from the value expected on the basis of classical mechanics, because a wave packet can be reflected both on its way up, and also on its way down the potential barrier separating the initial and final states. In fact, the transmission coefficient is, in many cases, a rapidly fluctuating function of the energy of the system. If the temperature distribution of the energy is sufficiently broad to cover several periods

of this fluctuation, an average transmission coefficient can be defined which nearly agrees with the classical value. For the crossing of a one-dimensional potential barrier, the quantum corrections are surprisingly small. In problems with several degrees of freedom, the transmission coefficient is affected by the interchange of translational and vibrational energy. However, if the vibrational motion is fast as compared with the motion along the reaction path, these degrees of freedom can be treated on a par with the electronic coordinates. In this case, the formulas of Eyring, with a mechanically sensible transmission coefficient, are satisfactory. On the whole, we conclude that quantum-mechanical considerations invalidate the transition state method to a much smaller extent than could be presumed and it is only in the consideration of the relative rates of reactions between isotopes and reactions at very low temperatures that these effects may be important.

I. THE TRANSITION STATE METHOD AND ITS VALIDITY

IT is well known that whenever classical mechanics is valid, the formula¹

$$k = (P_t/P_i)(\delta/\bar{v})^{-1}\gamma = P_t\bar{v}\gamma/(P_i\delta) \quad (1)$$

for the rate, k , of reactions with an activation energy is a direct consequence of statistical mechanics. In (1), P_t is the probability of the transition state in thermal equilibrium. The transition state is a strip of width δ in configuration space. This strip lies across the deepest saddle on the energy mountain separating the two regions in configuration space which correspond, respectively, to the initial and final state of the reaction. P_i is the probability of the system being in the initial state; $\bar{v}$ is the average velocity with which the configuration points cross the saddle; $\delta/\bar{v}$ their average time of sojourn in the transition state. Finally, γ expresses the proba-

bility that a system which crosses the saddle at complete thermal equilibrium actually originated in the initial state and will proceed to the final state to complete the chemical reaction.² The transmission coefficient, γ , is the only quantity appearing in Eq. (1) which cannot be evaluated by well-known methods of statistical mechanics—it can be estimated only from the general shape of the potential mountain.

There is no reaction with an activation energy in which classical mechanics is valid for all parts of the reacting system. However, it appears logical³ to replace the classical expressions for P_t and P_i in (1) by the corresponding quantum theoretical sums. The question as to what extent this is justifiable has been discussed repeatedly.⁴ One difficulty is that the notion of an activated

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¹ This equation was first proposed by A. Marcelin, *Ann. chim. phys.* (9)3, 120, 185 (1915). Subsequent treatments of this nature have been given by A. March, *Physik. Zeits.* 18, 53 (1917); Pelzer and Wigner, *Zeits. f. physik. Chemie* B15, 445 (1932); E. Wigner, *Zeits. f. physik. Chemie* B19, 203 (1932); H. Eyring, *J. Chem. Phys.* 3, 107 (1935); Evans and Polanyi, *Trans. Faraday Soc.* 31, 875 (1935).

² A more detailed explanation of (1) will be found, e.g. *Trans. Faraday Soc.* 34, 29 (1938). The reaction is considered "completed" when the distance between the reaction products becomes large compared with molecular dimensions. This has a clear sense only in gas reactions where the products separate at once to very large distances (of the order of the mean free path) if they begin to separate at all.

³ This was first done by E. Wigner (*Zeits. f. physik. Chemie* B19, 203 (1932)). It was also the method adopted by H. Eyring in developing his formulae of great generality.

⁴ See for example, E. Wigner, *Trans. Faraday Soc.* 34, 29 (1938).

or transition state is not strictly compatible with the laws of quantum mechanics. The transition state method can only be justified when the path of the reaction is sufficiently flat in the neighborhood of the transition state so that we can consider simultaneously the position and the velocity of the point representing the system in configuration space. In classical mechanics, δ is always considered to be so small that the potential is practically constant across the transition state. If δ is large, P_t and the average length of time that the system spends in the transition state become rather complicated functions of δ and (1) no longer applies. In quantum mechanics, it is necessary to take the width of the transition state sufficiently great to allow reducing the uncertainty in the velocity, Δv , to a small fraction of the average thermal velocity, $\bar{v}$. Certainly, if we are to apply Maxwell's formula, $\exp(-mv^2/2kT)$, for the probability of a system with a mass m to have the velocity v at the temperature T , the spread in the velocity, Δv , due to the uncertainty principle must be so small that it makes little difference whether we use the upper bound, $v + \Delta v/2$, or the lower bound, $v - \Delta v/2$, in the exponential. This means that $mv\Delta v$ must be small compared with kT . Putting $(kT/m)^{1/2}$ for v , this gives $\Delta v \ll (kT/m)^{1/2}$. We must take the transition state sufficiently wide so that this last relation can be satisfied. On the other hand, this width must be sufficiently small so that it is still a good approximation to take P_t proportional to δ . Thus, if the energy surface in the vicinity of the saddle has a curvature in the direction of the reaction path corresponding to the parabola, $V = V_0 - aX^2$, we must confine the activated state to a strip across the path which is so thin that $\exp(-aX^2/kT)$ shall be nearly unity all over the strip. This means that $\delta \ll (kT/a)^{1/2}$. These limits for the maximum values of Δv and of δ are only compatible with the quantum-mechanical uncertainty principle if:

$$\frac{h}{2\pi} < \delta m \Delta v \ll m(kT/a)^{1/2}(kT/m)^{1/2}.$$

Thus the transition state can only be defined when the curvature of the energy surface satisfies the condition:

$$(h/2\pi)(a/m)^{1/2} \ll kT. \quad (2)$$

Apart from an unimportant numerical factor, the left side of this equation is the zero point energy of a mass, m , in a potential, $+aX^2$. Thus, Eq. (1) and the transition states method can be justified only if this virtual zero point energy is smaller than kT . If Eq. (2) is satisfied and if the dependence of the potential on the coordinates along the strip does not change appreciably from one side of the strip to the other, the transition state can be defined sufficiently accurately for deriving Eq. (1) in spite of the limitations of quantum mechanics. Fortunately, this is fulfilled in many important cases.

The computation of γ presents another difficulty in the quantum-mechanical applications. As long as the reacting system behaves classically, γ can be estimated from simple geometrical and mechanical arguments. Usually the coordinates in configuration space are chosen so that the kinetic energy is the product of an effective mass and the sum of squares of the generalized velocities. In such coordinate systems, molecular collisions are kinematically equivalent to the rolling of a ball on a surface whose height is proportional to the potential energy of the corresponding molecular configuration. The question whether a given collision will result in a chemical reaction can be answered by examining the trajectory of the ball when it is shot in the corresponding manner. However, when quantum theory is used, the ball must be replaced by a wave packet. In some reactions γ will retain its classical value. In others, due primarily to the diffraction of the wave packets, it will have a different magnitude. Thus γ can be much smaller than unity in quantum-mechanical systems even for an energy surface for which a mechanical picture does not indicate an appreciable probability of reflection. Consequently, the use of Eq. (1) with seemingly reasonable values of γ can lead to erroneous rates even when the condition of (2) is fulfilled.

If the condition (2) is not fulfilled, (1) must certainly be corrected to take into account the quantum-mechanical penetration of the potential barrier.⁵ But it may not be possible to use a

⁵ R. E. Langer; Born and Franck, C. Eckart, Phys. Rev. **35**, 1303 (1930); R. P. Bell, Proc. Roy. Soc. **A139**, 466 (1933). That (2) is the condition for the absence of an appreciable amount of tunneling is evident also from the

penetration factor in any simple way to correct for the deviations from classical theory. If Eq. (2) is not fulfilled, a reflection after the crossing of the saddle (which is usually taken into account by choosing an appropriate value of γ) cannot be distinguished from a failure to cross the saddle. Thus it would no longer be possible to define a transmission coefficient.

II. THE TRANSMISSION COEFFICIENT

A. We shall first carry out a rather formal consideration concerning transmission coefficients, which will be based on the assumption that there is a definite probability for a system, which has crossed the transition state, to be reflected back to the transition state. This probability will be denoted by ρ_i and ρ_f for crossings from left to right and right to left, respectively. It will be assumed that these probabilities are equal for systems originating from the two sides of the transition state and are independent of the number of times the system has already crossed the transition state.

The value of the transmission coefficient, γ , depends of course on the exact nature of the reaction surface. It is unity if all systems cross the transition state only once in their passage from the initial to the final state or from the final to the initial state. In the general case, some of the systems⁶ which cross the transition state will cross it again in the other direction. This may happen repeatedly before the atoms of the system finally separate, either to form the same molecules which originally collided, or else to form the molecules corresponding to the completed reaction. Thus some trajectories which cross the transition state from left to right will not lead to a chemical reaction and others may require many crossings before the reaction is completed.

At complete thermal equilibrium, half of the systems in the transition state are moving from left to right and the other half from right to left.

formula for the first quantum correction to (1). (Cf. reference 3.)

⁶ The term "system" is used here also for the point in configuration space which corresponds to the position of the atoms forming the system. The left side of the transition state is supposed to correspond to the atoms forming the molecules of the reacting substances, the right side corresponds to the reaction product.

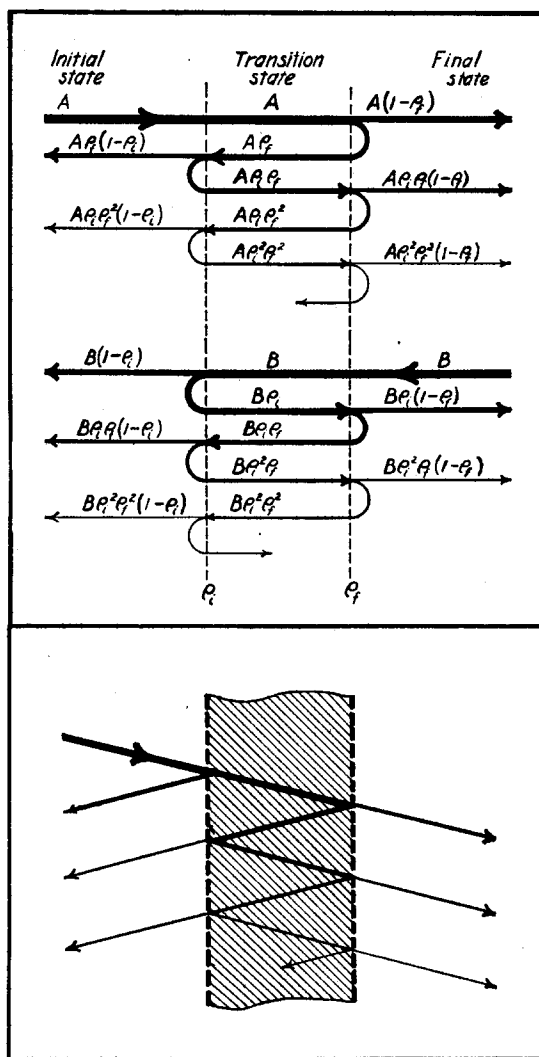


FIG. 1. The upper drawing shows the reflection of the systems on the two sides of the transition state. This is similar to the reflection of a beam of light at the air-glass and glass-air interfaces in passing through the plate glass shown in the lower figure.

The transmission coefficient, γ , is the fraction of the systems moving from left to right at complete thermal equilibrium which originally came from the initial state and which will proceed to the final state without first returning to the transition state. At complete thermal equilibrium, A systems arrive, in unit time, at the transition state directly from the initial state and B systems come from the final state. Fig. 1 shows schematically the flux of the different types of systems through the transition region. By simple addition, we see that the total number of systems

crossing the transition state from left to right in unit time is

$$N_{l \rightarrow r} = A(1 + \rho_i \rho_f + \rho_i^2 \rho_f^2 + \dots) + B \rho_i (1 + \rho_i \rho_f + \rho_i^2 \rho_f^2 + \dots) \\ = (A + B \rho_i)(1 - \rho_i \rho_f)^{-1} \quad (3)$$

and the number of those crossing it from right to left is

$$N_{r \rightarrow l} = A \rho_f (1 + \rho_i \rho_f + \rho_i^2 \rho_f^2 + \dots) + B(1 + \rho_i \rho_f + \rho_i^2 \rho_f^2 + \dots) \\ = (A \rho_f + B)(1 - \rho_i \rho_f)^{-1}. \quad (4)$$

At equilibrium $N_{l \rightarrow r} = N_{r \rightarrow l}$, and hence

$$B = A(1 - \rho_f)/(1 - \rho_i). \quad (5)$$

Substituting this expression for B into Eq. (3),

$$N_{l \rightarrow r} = A/(1 - \rho_i). \quad (6)$$

However, the number of systems which have originated in the initial state and which proceed to the final state is seen to be

$$N_{i \rightarrow f} = A(1 - \rho_f)(1 + \rho_i \rho_f + \rho_i^2 \rho_f^2 + \dots) \\ = A(1 - \rho_f)/(1 - \rho_i \rho_f). \quad (7)$$

The transmission coefficient is the ratio of $N_{i \rightarrow f}$ to $N_{l \rightarrow r}$

$$\gamma = N_{i \rightarrow f}/N_{l \rightarrow r} = (1 - \rho_i)(1 - \rho_f)/(1 - \rho_i \rho_f). \quad (8)$$

The assumption made at the beginning of this section that the systems crossing the transition state from left to right have the same probability for being reflected no matter whether they originated in the initial or in the final state (or how many times they have already crossed the transition state), is not always justified. We shall see in Section 3 that for a one-dimensional barrier, the transmission coefficient for any one value of the energy does not necessarily satisfy Eq. (8). However, the average transmission coefficient for many systems with slightly different energies will agree with the above expression.

We can compare the transmission of a wave packet through a potential barrier, as considered in this section, with the passage of a beam of light through a piece of plate glass partially silvered on both sides. (See lower part of Fig. 1.) Light of one particular frequency shows a very complicated diffraction pattern. If the direction of the beam is sharply defined, the amount of

light passing through the glass depends on the exact wave-length. For light of a range of frequencies, the transmission no longer depends critically on the wave-length and will be given by (8) (provided that the thickness of the plate is large compared with the wave-length). From this analogy we see that Eq. (8) can apply for the average transmission coefficient of an assembly of states in thermal distribution, not, however, for the transmission coefficient of a single quantum state. The example in Section 3 should make this more clear.

B. In classical theory, for low temperatures, ρ_i and ρ_f approach zero as it is very improbable that a system with barely enough energy to cross the activated state will find its way back.⁴ For higher temperatures, the reflection coefficients may increase. This must be expected, in particular, if the vibrations in the transition state are less stiff than in the initial state, i.e., for the exceptionally fast reactions.

We shall consider an energy surface on which the energy and vibrational frequency change within a very short distance along the reaction path. The abruptness of the energy change tends to make γ small; while the straightness of the reaction path tends to make it large. We shall suppose that the energy surface at the initial state has a smaller curvature perpendicular to the reaction path than at the final state, but that the minimum energy of the initial state is higher than that of the final state. Fig. 2 shows such an energy surface. All the systems of low energy pass from the initial to the final state, but part of the systems of high energy are reflected. If the potential energy is $V_i = A + a_i x^2$ for the initial and $V_f = a_f x^2$ for the final state, only those systems can be reflected which lie outside the point of intersection, $x' = A^{1/2}/(a_f - a_i)^{1/2}$. This is a negligibly small fraction of the systems at low temperatures and hence $\gamma = 1$. The reflection plays an important role only at temperatures so high that $kT > a_i x'^2$. This is equivalent to the condition:

$$A/kT < a_f/a_i - 1 = (\nu_f/\nu_i)^2 - 1, \quad (9)$$

where ν_i and ν_f are the vibrational frequencies in initial and final states.

The existence of a limiting temperature, above which γ becomes small, will always be true

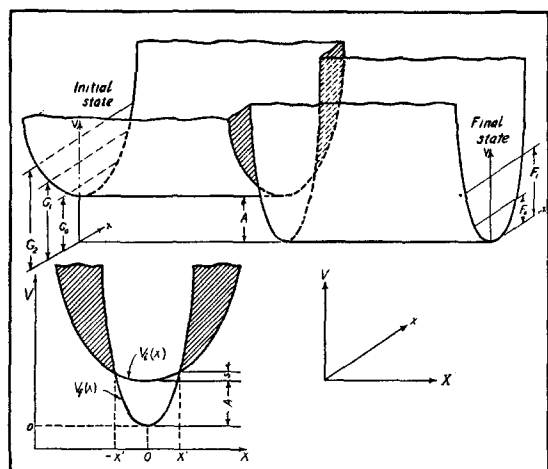


FIG. 2. Potential energy surface illustrating the variation of the transmission coefficient with temperature. From classical mechanical considerations, only those systems can be reflected which have sufficient energy to reach the baffle. The coordinate axes refer to the upper figure.

whenever the frequencies in the transition state are lower than in the initial state. The change in energy between the initial and the final state does not have to take place abruptly nor does the reaction path have to be straight. This becomes evident if we compare the rate of the reverse reaction, according to Eq. (1), with the total number of collisions in which the energy of the colliding particles is greater than A . Clearly, this latter quantity gives an upper limit for the rate.⁷ The number of collisions involving energies greater than A is

$$(kT/2\pi m)^{1/2} (A/kT + 1) \exp(-A/kT). \quad (10)$$

This must be larger than the reaction rate according to Eq. (1):

$$k = \gamma(\nu_f/\nu_i)(kT/2\pi m)^{1/2} \exp(-A/kT). \quad (11)$$

Thus γ must be less than unity if

$$A/kT < \nu_f/\nu_i - 1. \quad (12)$$

The limiting temperature for a transmission coefficient of unity as given by Eq. (12) is about right if the reaction path is straight and the transverse vibration frequencies change slowly along the reaction path. The temperature given by (12) is much higher than that of Eq. (9). This corresponds to the fact that γ is larger for

⁷ K. F. Herzfeld, *Zeits. f. Physik* **8**, 132 (1922); M. Polanyi, *Zeits. f. Physik* **1**, 337 (1920).

energy surfaces having smooth rather than abrupt changes. In actual chemical reactions, γ will depart from unity somewhere between these two temperatures. Thus, in general, the transmission coefficient falls below 1 at high temperatures for classical mechanical reasons. We shall see in the next section that at low temperatures, where γ should be 1 according to classical mechanics, it is also smaller than 1 for quantum-mechanical reasons.

III. ONE-DIMENSIONAL RATE PROBLEMS

The simplest examples of reaction rate involve motion in only one dimension. If the activated state method applies, we can divide the calculation of γ into the two easier problems of determining the reflection coefficients ρ_f and ρ_i which refer to the passage from the activated state to the final and to the initial states, respectively. For one-dimensional problems these reflection coefficients are, of course, zero in classical mechanics.

In the special case of Eckart's potential functions,

$$V(X) = A(1 + \exp(2\pi X/l))^{-1}, \quad (13)$$

decreases from A to 0, as X increases from large negative values (initial state) to large positive values (final state). The energy drop takes place around $X=0$ in a distance roughly equal to $l/2$ (Fig. 3). The reflection coefficient corresponding to the motion from negative to positive X is:

$$\rho = \left[\frac{\exp(2\pi l(p-q)/h) - 1}{\exp(2\pi l(p+q)/h) - 1} \right]^2 \exp(4\pi lq/h), \quad (14)$$

where $q = (2m(E-A))^{1/2}$ and $p = (2mE)^{1/2}$ are the momenta of the system before and after the potential drop. As this drop becomes abrupt ($l \rightarrow 0$), the reflection coefficient becomes

$$\rho = (p-q)^2/(p+q)^2. \quad (15)$$

The explanation of this reflection is similar to that for the reflection of light at a glass to air interface. In quantum mechanics the dynamical system is represented by a wave packet with a wave-length which decreases from h/q to h/p as it passes across the potential drop. The energy change can therefore be expressed as a change in

the index of refraction and the analogy with light is complete.

In Fig. 4 we have plotted the reflection coefficient for an H atom when passing across abrupt, steep and gradual Eckart potential drops of 10 kcal. The gradual drop, $l=1.0\text{\AA}$, is still somewhat steeper than those which occur in the majority of chemical reactions. We see that the reflection coefficient approaches zero rapidly as the energy of the system is increased. One will expect, therefore, that the quantum corrections in one-dimensional problems are important only for reactions at very low temperatures involving light molecules or for isotope separations where slight differences in rates are significant.

Strictly speaking, this conclusion holds only if the barrier is so wide that (2) is satisfied, and consequently the method of Section 2 can be used to express the transmission coefficient in terms of the reflection coefficients. Otherwise the transmission must be computed by quantum-mechanical methods. For the sake of concreteness, let us consider a potential barrier with abrupt energy changes (Fig. 5). We denote by p_i , q and p_f the momenta of the system in the initial, transition, and final states. Substituting the reflection coefficient (15) into (8), we would obtain for the transmission coefficient of the barrier:

$$\bar{\gamma} = 4qp_i p_f / (q^2 + p_i p_f)(p_i + p_f). \quad (16)$$

In this expression there is no dependence of the transmission coefficient on the width of the barrier, d .

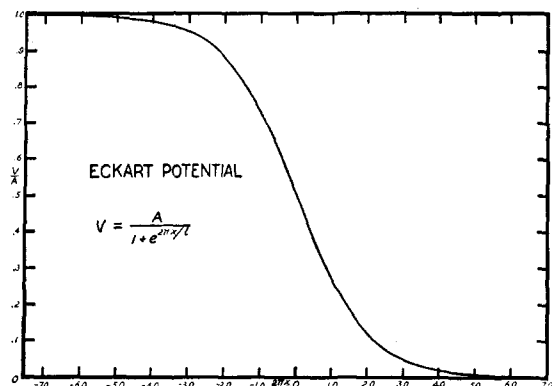


FIG. 3. A special case of the Eckart potential gives a function which changes smoothly from one constant value to another.

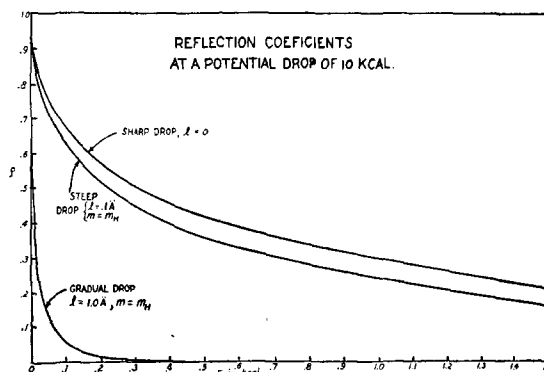


FIG. 4. Probability of reflection at abrupt, steep and gradual potential energy drops as a function of the energy of the system.

One can compute the exact value of the transmission coefficient by fitting together the wave function and its derivative at the discontinuities of the potential energy. One obtains in this way

$$\gamma = \frac{4q^2 p_i p_f}{q^2 (p_i + p_f)^2 \cos^2 \varphi + (q^2 + p_i p_f)^2 \sin^2 \varphi}, \quad (17)$$

where $\varphi = 2\pi qd/h$. This is evidently different from (16). If the width of the barrier is not large compared with the wave-length, quantum mechanics prevents us from even defining a transition state. If the barrier is wide enough to justify the activated state method, the exact transmission coefficient of Eq. (17) shows rapid fluctuations with the energy. Averaging γ over a small energy range (which corresponds under these conditions to averaging over φ) one obtains an averaged transmission coefficient, which turns out to be equal to the transmission coefficient of Eq. (16):

$$\bar{\gamma} = \frac{1}{2\pi} \int_0^{2\pi} \gamma d\varphi. \quad (18)$$

It is only in this sense that Eq. (8) is valid. For any particular energy, the waves reflected at the two discontinuities of the potential interfere with each other to aid or to hinder the transmission. The effect of this interference averages out, however, for reasonably large energy ranges.

Figure 6 shows γ and $\bar{\gamma}$ plotted as functions of the energy of the impinging particles. Here $A_1 = A_2 = A = 10$ kcal. and $d = 1.0\text{\AA}$. The exact

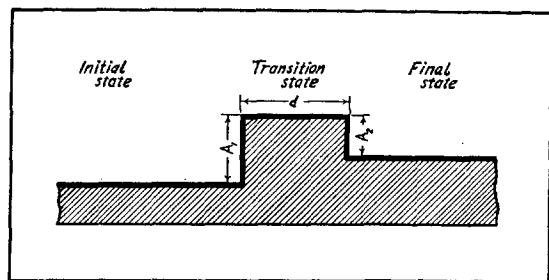


FIG. 5. One-dimensional energy barrier with abrupt changes in the potential.

transmission coefficient, γ , oscillates about $\bar{\gamma}$. The oscillation of the transmission coefficient becomes quicker with increasing mass of the particles. It is interesting to note that the maximum in the transmission coefficient occurs for deuterium before it occurs for hydrogen so that at an energy of 10.2 kcal., the transmission coefficient for the former is about five times greater than for the latter. The situation is reversed, however, at 10.5 kcal.

It seems tempting, at first, to try to utilize the large ratios of transmission coefficients, as shown in Fig. 6, for the separation of isotopes. This is difficult, however, for several reasons. First, because the energy of the reacting systems is not restricted to one single value in actual experiments but covers—according to the Maxwell-Boltzmann distribution formula—a region of appreciable width so that the fluctuations shown in Fig. 6 average out to a large extent. Furthermore, the ratio of transmission coefficients fluctuates not only as a function of the energy but in many dimensional problems also as a function of the other parameters of the problem. As these parameters also cover a range of values

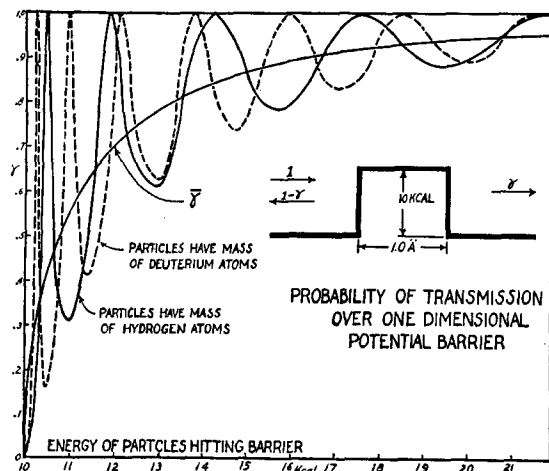


FIG. 6. Transmission coefficients for H and for D atoms passing over a one-dimensional potential energy barrier.

for the reacting systems, the fluctuations tend to average out to an even larger extent. Finally, the fluctuations are as pronounced as in Fig. 6 only if the top of the barrier has a flat portion, which is not very much shorter than the wavelength.

On the other hand, one can, in principle, always obtain large separations at very low temperatures, where "tunneling" becomes important. In quantum mechanics, systems with less energy than the activation energy may still react although only with a small probability.⁸ The transmission coefficient is still given by (17) where now q is imaginary.

In unit time the number of molecules with an energy between E and $E+dE$ which hit the barrier is proportional to $\exp(-E/kT)dE$. The ratio of the actual rate, k , to the classical rate, k_{class} , is then:

$$\frac{k}{k_{\text{class}}} = \frac{\int_0^A \gamma(E) \exp(-E/kT) dE + \int_A^\infty \gamma(E) \exp(-E/kT) dE}{\int_A^\infty \exp(-E/kT) dE} \quad (19)$$

$$= \frac{k_{\text{through}}}{k_{\text{class}}} + \frac{k_{\text{over}}}{k_{\text{class}}}.$$

⁸ Similar calculations were made before particularly by R. P. Bell, J. Chem. Phys. 2, 164 (1934); Proc. Roy. Soc. A139, 466 (1933); H. Eyring and A. Sherman, J. Chem. Phys. 1, 345 (1933); C. E. H. Banri and G. Ogden, Trans. Faraday Soc. 30, 432 (1934).

Figure 7 shows the relative number of D atoms which react at 126°K as a function of their energy. There are two peaks in this curve. One corresponds to the reactions, k_{through} , due to penetration of the barrier and the other, k_{over} , corresponds to passage over the barrier. At a slightly higher temperature the penetration is unimportant; at a slightly lower temperature the passage over the barrier is unimportant. At high temperatures, where classical theory is valid, the rate of the crossing of the barrier is $2\frac{1}{2}$ times higher for hydrogen than for deuterium, owing to the higher velocity of the former. Table I shows the rates of crossing, for H and for D, relative to the classical rate of H at 252°. Here k_{trans} is the reaction rate which would be predicted on the basis of the transition state method using the quantum-mechanical value for the reflection coefficients, and Eq. (8)

$$\frac{k_{\text{trans}}}{k_{\text{class}}} = \frac{\int_A^\infty \bar{\gamma}(E) \exp(-E/kT) dE}{\int_A^\infty \exp(-E/kT) dE}$$

At room temperatures, the penetration of this barrier is unimportant; the separation of the isotopes is small; and the transition state method is satisfactory. At low temperatures the transition state method is inapplicable since most reactions occur by penetration of the barrier. Here the reaction rates are thousands of times faster for the lighter isotope. The above table is in agreement with the calculations of Bell⁸ who used an Eckart type of potential barrier. From this table we can make the following observations:⁴ At very low temperatures an experimenter would find that a reaction of this nature has an anomalously small steric factor and shows practically no "activation energy." (But at temperatures so low that k_{through} is greater than k_{over} , the rates are so small that they can hardly be measured.) At high temperatures he would obtain an activation energy agreeing with his classical expectations but the steric factor would still be small by a factor of two or three.

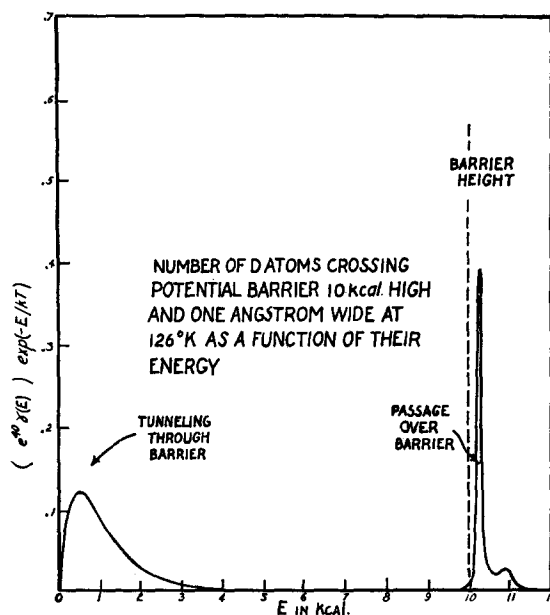


FIG. 7. Relative number of D atoms which succeed in crossing a potential energy barrier 10 kcal. high and 1 Å wide plotted as a function of their energy. The systems approaching the barrier have the energy distribution corresponding to 126°K.

IV. THE SIMPLEST MANY-DIMENSIONAL RATE PROBLEM—EYRING'S FORMULATION

Reaction rate problems in many dimensions are more complicated than the one-dimensional examples principally because of the interchange of energy between the various degrees of freedom. In this section we consider the simplest of all many-dimensional problems—a straight reaction path along which the motion is slow and perpendicular to which the motion is fast.

If we designate by X the coordinate along the reaction path and x the coordinate perpendicular to the reaction path (considering only one such coordinate x for the sake of simplicity), the potential energy of the system is $U(X, x)$. We

TABLE I.

	T	$\frac{k_{\text{THROUGH}}}{k_{\text{CLASS}}}$	$\frac{k_{\text{OVER}}}{k_{\text{CLASS}}}$	k	k_{CLASS}	k_{TRANS}
H	84°	$1.7 \cdot 10^{13}$	0.15	$7.1 \cdot 10^{-5}$	$4.3 \cdot 10^{-18}$	$3.2 \cdot 10^{-19}$
	126°	$7.2 \cdot 10^4$	0.22	$1.5 \cdot 10^{-4}$	$2.1 \cdot 10^{-9}$	$2.9 \cdot 10^{-10}$
	252°	0.014	0.35	0.364	1.0	0.39
D	84°	$1.5 \cdot 10^8$	0.22	$4.6 \cdot 10^{-10}$	$3.0 \cdot 10^{-18}$	$2.3 \cdot 10^{-19}$
	126°	0.76	0.26	$1.5 \cdot 10^{-9}$	$1.5 \cdot 10^{-9}$	$2.0 \cdot 10^{-10}$
	252°	0.0023	0.37	0.26	0.707	0.28

denote by x_0 the point x at which $U(X, x)$ assumes its minimum value for a given X . Obviously x_0 is a function of X . $U(X, x)$, as a function of x has approximately the shape of a parabola in the neighborhood of $x=x_0$. If the motion along the reaction path is so slow that the system can vibrate many times in the x direction before it proceeds so far in the direction of X that the curvature of the parabola is changed considerably, the motion perpendicular to the reaction path can be considered as simple harmonic with the frequency $\nu(X)$. The potential energy which is responsible for the motion of the slow coordinate X is then

$$V_n(X) = U(X, x_0) + (n + \frac{1}{2})h\nu(X). \quad (20)$$

Here n is the quantum number of the vibration along x . If $V_n(X)$ is a slowly varying function of X , the motion along the reaction path will obey the classical equations of motion.

It is essential in the derivation of Eyring's formulae¹ that the motion in the fast and in the electronic coordinates be adiabatic in the sense that neither the electronic quantum states nor n undergo any changes. We can use the ordinary methods of perturbation theory to obtain the corresponding condition on the velocity, $v=dX/dt$, and on the change in the vibration frequency along the reaction path, $d\nu(X)/dX$,

$$d\nu/dt = v d\nu(X)/dX \ll \nu^2. \quad (21)$$

Here v can be much greater than the velocity corresponding to the average thermal energy since it must be fast enough to permit the crossing of the potential barrier. Thus (21) is more stringent than for the case of motion with ordinary thermal energies of the order of kT .

Under the above conditions, the motion of a system with the vibrational quantum number n will obey classical mechanics and will be governed by the potential (20). Therefore, the number of systems with this vibrational quantum number which react in unit time will be given by (1) into which

$$P_{in}^* = \delta \exp(-V_n(X_0)/kT) \quad (22)$$

must be substituted for P_i , where $V_n(X_0)$ is the highest value of $V_n(X)$, corresponding to the activated state.

The total reaction rate is the sum of the rates of all systems in all possible vibrational states. This is

$$\sum_n P_{in} \bar{v} \gamma / \delta P_i = P_i \bar{v} \gamma / \delta P_i, \quad (23)$$

where

$$P_i = \sum_n P_{in} = \delta \exp(-V_0(X_0)/kT) \times (1 - \exp(-h\nu(X_0)/kT))^{-1}. \quad (24)$$

Thus under these assumptions we are led back to Eq. (1) except that the quantum theoretical probability appears in it, instead of the classical one.

It may be worth while to remark in this connection that if the reflection coefficients are zero and the molecules are distributed before the reaction according to thermal equilibrium, the thermal equilibrium will be maintained throughout the course of the reaction. Consider each vibrational state as forming a separate energy surface with respect to the motion along the reaction path. On the surface characterized by the vibrational quantum number n , the density at any point, X , is given by the barometric formula, $C_n \exp((V_n(\text{initial}) - V_n(X))/kT)$. The condition that the systems be at thermal equilibrium initially is that

$$C_n = C \exp((V_0(\text{initial}) - V_n(\text{initial}))/kT).$$

From this it follows that the ratio of the density on the n th to that on the zeroth vibrational state is $\exp((V_0(X) - V_n(X))/kT)$ for any value of X .

The consideration of this section shows that it is easy to justify Eyring's formulae with a mechanically sensible transmission coefficient if the coordinates can be divided into some which involve slow motion and others which involve fast motion. The slow motion can be described by classical mechanics. The fast motion can be treated on a par with the motion of the electrons: the corresponding quantum numbers undergo no changes during the reaction. This condition restricts the consideration of this section to potential surfaces where the frequencies along the reaction path change slowly.

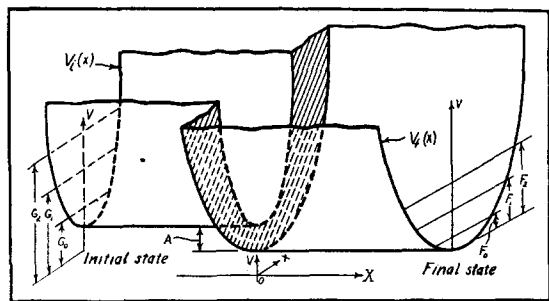


FIG. 8. Two-dimensional potential energy trough with straight reaction path and an abrupt change in the potential. Here the G 's are the vibrational energy levels in the initial state and the F 's in the final state.

V. ABRUPT CHANGE OF POTENTIAL ENERGY ON TWO-DIMENSIONAL SURFACES

The difficulty of solving partial differential equations restricts the number of many-dimensional rate problems which we can treat accurately. In this section we consider examples in which the reaction path is straight and the change in the potential occurs in a very short distance. It would also be desirable to consider the effect of curvature in the reaction path. The potential energy surfaces are constructed so that the Schrödinger equation is separable in each of a number of regions. In each region, the wave functions are obtained as solutions of ordinary (one-dimensional) differential equations. The wave functions are then pieced together so that the functions and their derivatives are continuous at the boundaries of every two regions. This is carried out by matrix methods and the complete mathematical treatment is given in the appendix.

Let us consider an energy surface similar to that shown in Fig. 8. For negative X , the potential energy is $V_i(X) = A + a_i x^2$; for positive X , the potential energy is $V_f(X) = a_f x^2$. This surface is the two-dimensional analog of the potential drop of Section 3. Eq. (8) enables us to obtain the average transmission coefficient from the reflection coefficients at two energy drops similar to that of Fig. 8. The case in which the energy of the system is large compared with the vibrational energy of any of the levels in the final state to which the waves are likely to be transmitted is particularly easy to treat.

The vibrational quantum number of the reflected systems can differ, to the approximation

used in the appendix, only by 0 or ± 2 from the vibrational quantum number of the incident systems. Eqs. (36) and (40) of the Appendix show that even the change by ± 2 is quite improbable as ϵ is always a rather small quantity. The transmitted waves can have a vibrational quantum number differing from that in the initial state by an even number. In Fig. 9, we have plotted the probabilities of reflection and transmission into states with different vibrational quantum numbers. One sees by comparison with Fig. 4, that the reflection coefficient is rather similar to that of a system with equal translational energy in a one-dimensional problem. One sees, furthermore, that the vibrational quantum number has a marked persistence even in this case of an abrupt potential change. This means that the vibrational energy increases if the frequency is larger after the drop, and decreases in the opposite case. The changes in the vibrational frequency, inasmuch as they occur, have a tendency to counteract this. Thus the vibrational quantum number is more likely to decrease if the vibration is stiffer after the drop and is more likely to increase in the opposite case. A more detailed discussion is given at the end of the Appendix.

The persistence of the vibrational quantum number is doubtless in close connection with the slowness with which the equilibrium distribution of vibrations is established. At higher vibrational quantum numbers changes in the vibrational quantum number become more probable which is also in qualitative agreement with experiment.⁹

For the study of reaction rates, the reflection coefficients are most important. It is seen that in the case investigated here—large kinetic energy in the final state—these reflection coefficients are not increased over the value they assume in a similar one-dimensional problem. This leads to the conclusion that the transmission coefficient is even in this case not far from unity and Eyring's well-known formulae remain applicable. It would be desirable to obtain a more general solution of the problem considered in this section than that given in the Appendix and we intend to return to this question at a later time.

⁹ Cf. F. Patat, *Zeits. f. Elektrochem.* **42**, 265 (1936).

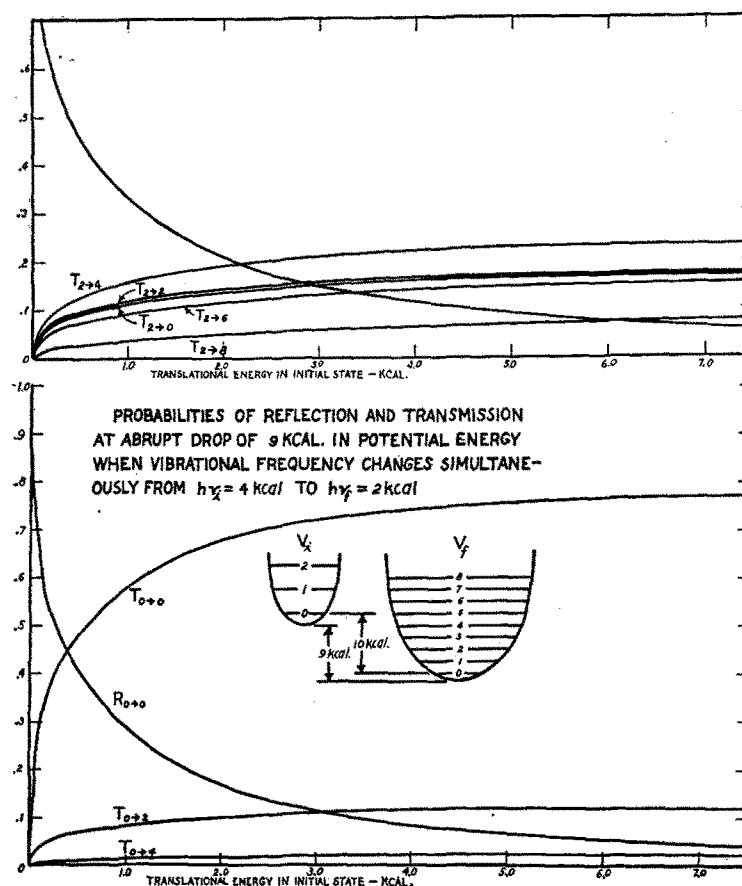


FIG. 9. Probabilities of reflection and of transmission at an abrupt potential drop of 9 kcal. when the vibrational frequency changes simultaneously from $h\nu_i = 4$ kcal. to $h\nu_f = 2$ kcal. The lower curves are for systems initially in the zeroth vibrational state; the upper curves are for systems initially in the 2nd vibrational state.

APPENDIX

Calculation of transmission and reflection coefficients for many dimensional problems

Let us denote by X the coordinate along the reaction path and by x the coordinate (or coordinates) perpendicular to it. For $X < 0$ the potential energy is $V_i(x)$; for $X > 0$ it is $V_f(x)$. The characteristic functions and characteristic values of the Schrödinger equation $H_i\psi = I\psi$ with the potential $V_i(x)$ shall be $\psi_k(x)$ and I_k , those of the Schrödinger equation $H_f\varphi = F\varphi$ with the potential $V_f(x)$ will be $\varphi_k(x)$ and F_k . Greek letters as subscripts refer to the initial state, roman letters to the final state. Expanding the φ in terms of the ψ we obtain

$$\varphi_k(x) = \sum_{\lambda} u_{k\lambda} \psi_{\lambda}(x) \quad (26)$$

and u is a unitary matrix

$$(u^{-1})_{\lambda k} = u_{k\lambda}^*. \quad (27)$$

We shall consider an incident wave from the left having the form $\psi_k(x) \exp(2i\pi q_k X/h)$. It will give rise to a reflected beam in which the atoms have all possible quantum numbers λ for their motion in the x direction. Thus the wave function for $X < 0$ becomes

$$\psi_k(x) \exp(2i\pi q_k X/h) + \sum_{\lambda} R_{\lambda k} \psi_{\lambda}(x) \exp(-2i\pi q_{\lambda} X/h) \quad (28)$$

with

$$q_k^2/2m + I_k = E, \quad (28a)$$

where E is the total energy of the system. For

$X > 0$ we have only an outgoing wave

$$\sum_k T_{\kappa k} \varphi_k(x) \exp(2\pi i p_k X/h) \\ = \sum_{k\lambda} T_{\kappa k} u_{k\lambda} \psi_\lambda(x) \exp(2\pi i p_k X/h), \quad (29)$$

where again

$$p_k^2/2m + F_k = E. \quad (29a)$$

Naturally, the p_κ and q_k with high quantum numbers κ and k will be imaginary. The corresponding terms represent local disturbances in the neighborhood of the discontinuity. The imaginary p and q , must be, if divided by i , positive. Terms with imaginary momenta of the opposite sign of p or q would correspond to waves which have progressively greater amplitudes as we depart from the discontinuity. The total number of systems with quantum number λ reflected in unit time by the discontinuity is $R_{\kappa \rightarrow \lambda} = |R_{\kappa \lambda}|^2 q_\lambda / m$, if q_λ is real (there is of course no reflected wave with imaginary q_λ). Similarly, the total number of systems transmitted in unit time with the quantum number k into the final states is $T_{\kappa \rightarrow k} = |T_{\kappa k}|^2 p_k / m$ if p_k is real, zero if p_k is imaginary. There are q_κ / m systems incident in unit time.¹⁰

Both the wave function and its derivative must be continuous at $X=0$. Comparing the coefficients of $\psi_\lambda(x)$ of Eqs. (28) and (29) and of their derivatives this gives:

$$\delta_{\kappa \lambda} + R_{\kappa \lambda} = \sum_k T_{\kappa k} u_{k\lambda}, \quad (30)$$

$$\delta_{\kappa \lambda} q_\kappa - R_{\kappa \lambda} q_\lambda = \sum_k T_{\kappa k} u_{k\lambda} p_k. \quad (30a)$$

The formal solution of these equations can be obtained by matrix algebra. We can write for (30) and (30a)

$$1 + R = T u, \quad (31)$$

$$q - R q = T p u. \quad (31a)$$

Here 1 is the unit matrix, p and q diagonal matrices while R , T and u are general matrices. We have, from (31) and (31a), $T = (1 + R)u^{-1} = (1 - R)qu^{-1}p^{-1}$ and hence, with $\zeta = qu^{-1}p^{-1}u$,

$$R = (\zeta - 1)(\zeta + 1)^{-1}; \quad T = 2\zeta(\zeta + 1)^{-1}u^{-1}. \quad (32)$$

¹⁰ The matrix method for problems of atomic scattering has been initiated by J. A. Wheeler, Phys. Rev. 52, 1107 (1937).

We can use (32) to verify that the total number of incident systems is equal to the number of reflected and transmitted systems. The reflection and transmission matrices $\bar{R}$ and $\bar{T}$ for the problem in which the potential is V_f in the initial and V_i in the final state, can be calculated from R and T

$$\bar{R} = -uRu^{-1}; \quad \bar{T} = u(1 - R). \quad (32a)$$

Naturally, (32) represents only a formal solution of our problem, and the matrix elements of R and T still remain to be calculated. We shall obtain them for the special case in which the total energy E is very large compared to the energies F_k of the levels to which there is an appreciable transition. This means that in the final state the translational energy $p_k^2/2m \gg F_k$ will be large compared with the vibrational energy, although this need not be true for the initial state.

In this case $F_k \ll E$ and¹¹ we can write, denoting by f the diagonal matrix with the diagonal elements F_k ,

$$\zeta = qu^{-1}p^{-1}u \\ = qu^{-1}(2mE)^{-1/2}(1 + f/2E)u = \zeta_0 + \zeta_1, \quad (34)$$

where we have with $P = (2mE)^{1/2}$

$$\zeta_0 = P^{-1}q; \quad \zeta_1 = \frac{1}{2}P^{-1}E^{-1}qu^{-1}fu \quad (34a)$$

and ζ_1 is small compared with ζ_0 . Now we can write, up to terms of second order in ζ_1

$$R = (\zeta_0 - 1 + \zeta_1)(\zeta_0 + 1 + \zeta_1)^{-1} \\ = (\zeta_0 - 1 + \zeta_1)(\zeta_0 + 1)^{-1} [(\zeta_0 + 1 + \zeta_1)(\zeta_0 + 1)^{-1}]^{-1} \\ = (\zeta_0 - 1 + \zeta_1)(\zeta_0 + 1)^{-1} [1 - \zeta_1(\zeta_0 + 1)^{-1}] \\ = (\zeta_0 - 1)(\zeta_0 + 1)^{-1} + 2(\zeta_0 + 1)^{-1}\zeta_1(\zeta_0 + 1)^{-1}. \quad (35)$$

Hence

$$R_{\kappa \lambda} = \frac{q_\kappa - P}{q_\kappa + P} \delta_{\kappa \lambda} + \frac{PE^{-1}q_\kappa F_{\kappa \lambda}}{(q_\kappa + P)(q_\lambda + P)}, \quad (36)$$

where

$$F_{\kappa \lambda} = \sum_k u_{\kappa k}^* F_k u_{k\lambda} = \int \psi_\kappa^* H_f \psi_\lambda.$$

For the amplitude of the wave reflected without any change in the vibrational quantum number,

¹¹ In this calculation, the zero of the energy is at the bottom of the potential curve $V_f(x)$. E and P are numbers, the other symbols matrices.

the expression

$$R_{\kappa\kappa} = (q_\kappa - p_\kappa')(q_\kappa + p_\kappa')^{-1}; \quad p_\kappa'^2/2m + F_{\kappa\kappa} = E \quad (37)$$

is correct up to second-order terms in F . This shows a marked analogy to (15) with the difference, however, that p_κ' enters instead of p_κ . The former is smaller than the latter, at least for $\kappa=1$, as $F_{11} > F_1$. Hence, the reflection without change of the vibrational quantum number is smaller in the two-dimensional case than the whole reflection in the one-dimensional case. In order to calculate the $F_{\kappa\lambda}$ we can note that

$$\begin{aligned} F_{\kappa\lambda} &= \int \psi_\kappa^* H_f \psi_\lambda = \int \psi_\kappa^* H_i \psi_\lambda - \int \psi_\kappa^* (V_i - V_f) \psi_\lambda \\ &= I_\kappa \delta_{\kappa\lambda} - (V_i - V_f)_{\kappa\lambda}. \end{aligned} \quad (38)$$

If V_i and V_f have both the potential of a simple harmonic oscillator,

$$V_i = A + 2\pi^2 m \nu_i^2 x^2; \quad V_f = 2\pi^2 m \nu_f^2 x^2, \quad (39)$$

(36) and (38) show that the vibrational quantum number can change only by 0 or ± 2 in our approximation. Then

$$\begin{aligned} (V_i - V_f)_{\kappa\kappa} &= A + \frac{1}{2} \epsilon \hbar \nu_i (\kappa + \frac{1}{2}), \\ (V_i - V_f)_{\kappa, \kappa+2} &= (V_i - V_f)_{\kappa+2, \kappa} \\ &= \frac{1}{2} \epsilon \hbar \nu_i (\kappa + 1)^{\frac{1}{2}} (\kappa + 2)^{\frac{1}{2}} \end{aligned} \quad (40)$$

with $\epsilon = (\nu_i^2 - \nu_f^2)/\nu_i^2$. There is no change whatever in the vibrational quantum number if the frequencies in initial and final states are equal. In general, the probability of a change increases with increasing vibrational energy in the initial state, and also with increasing translational energy as long as the latter remains small compared with the total drop A in the potential.¹²

¹² It should be remembered that the probability of reflection is $|R_{\kappa\lambda}|^2 q_\lambda / q_\kappa$, not $R_{\kappa\lambda}$.

For the transmission matrix we obtain from (31)

$$\begin{aligned} T_{\kappa\kappa} &= \sum_\lambda (\delta_{\kappa\lambda} + R_{\kappa\lambda}) u_{\kappa\lambda}^* \\ &= \frac{2q_\kappa}{q_\kappa + P} u_{\kappa\kappa}^* + \sum_\lambda \frac{P E^{-1} q_\kappa F_{\kappa\lambda} u_{\kappa\lambda}^*}{(q_\kappa + P)(q_\lambda + P)}. \end{aligned} \quad (41)$$

The second term contains the factor F/E and is much smaller than the first. We shall evaluate (41) also for the potentials (39). In this case $T_{\kappa\kappa}$ will be zero unless the change $(k - \kappa)$ of the vibrational quantum number is even. This is because both $F_{\kappa\lambda}$ and $u_{\kappa\lambda}$ vanish, unless the difference of their two indices is even. The first few u are

$$\begin{aligned} u_{00} &= (1 - \epsilon)^{\frac{1}{2}} (1 - \epsilon + \frac{1}{4} \epsilon^2)^{-\frac{1}{2}}, \\ u_{02} &= -u_{20} = 2^{-\frac{1}{2}} u_{00} \epsilon (2 - \epsilon)^{-1}, \\ u_{22} &= u_{00} (1 - \epsilon - \frac{1}{8} \epsilon^2) (1 - \epsilon + \frac{1}{4} \epsilon^2)^{-1}. \end{aligned} \quad (42)$$

If the change in vibrational frequency is not too large, ϵ will be small. Under these conditions, at least for low values of the indices, the diagonal elements of u are nearly 1. The off-diagonal elements are much smaller, as $u_{\kappa\kappa}$ contains the $\frac{1}{2}(k - \kappa)$ 'th power of ϵ . Thus the probability for a change in κ is small. For large values of the quantum numbers, the vibrational energy in the final state covers a wider range. In the classical region, it extends from the vibrational energy in the initial state I_v to $I_v(1 - \epsilon)$. Its mean value is still very nearly the same as if there was no change in vibrational quantum number: it is $I_v(1 - \frac{1}{2}\epsilon)$ instead of $I_v(1 - \epsilon)^{\frac{1}{2}}$.

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